

Jonathan Chan

+447453417448 | jonathan.chan0@protonmail.com | jonchan0.github.io

EDUCATION

University of Oxford, Merton College

DPhil in Medical Sciences (Bioinformatics/Cardiovascular Genomics)

Thesis: Multifactorial predictive modelling of disease risk and trajectory in hypertrophic cardiomyopathy.

Oxford, United Kingdom

2022 – Present

University of Oxford, St Hilda's College

MBiochem in Molecular and Cellular Biochemistry – 1st Class (Rank: 2nd of 97)

Oxford, United Kingdom

2018 – 2022

HONOURS

- Radcliffe Department of Medicine & Clarendon Fund DPhil Scholarship (~£175k), 2022
- Gibbs Prize for Proxime Accessit in Biochemistry (2nd in Cohort), 2022
- Department of Biochemistry Project Prize for Master's Research (3rd in Cohort), 2022

RESEARCH EXPERIENCE

University of Oxford Radcliffe Department of Medicine, Oxford, United Kingdom

Doctoral Student (Prof. Hugh Watkins & Prof. Anuj Goel), Sep 2022 – Present

- Integrated genomic, proteomic, clinical and imaging data across patient (HCM Registry) & population (UK Biobank, All of Us) datasets to develop multifactorial risk prediction models for hypertrophic cardiomyopathy (HCM) patient stratification.
- Discovered novel biomarkers of HCM (ANGPT2, LTBP2) with diagnostic and prognostic utility via population-scale plasma proteomic analyses (UK Biobank, All of Us, HCM Registry). First-author publication in *Circulation: Precision and Genomic Medicine*.
- Applied statistical genetic approaches (GWAS, fine-mapping, Mendelian randomisation, colocalisation) to identify causal associations between HCM and disease-associated risk factors/biomarkers.
- Conducted genotype-phenotype correlation analyses in Mendelian HCM using clinical and imaging data with systematic review/meta-analysis to identify patient subgroups with greater disease severity.

Dart Biosciences Ltd, London, United Kingdom

Computational Biologist (Part-Time), Oct 2024 – Present

- Integrated multi-omic datasets (scRNA-seq/ATAC-seq) datasets to identify activation-dependent genes and cis-regulatory elements, contributing to provisional patent submission.
- Benchmarked state-of-the-art single-cell transcriptome → surfaceome (CITE-seq) prediction models, producing a data-based go/no-go recommendation.
- Rapidly prototyped MVP for cell-specific marker gene identification (*cellxgene*, *scVI*, *cellPLM*), ultimately driving a strategic decision by identifying commercial viability constraints.
- Leading the *in silico* design of synthetic cis-regulatory elements (CREs) for cell-type specific *in vivo* payload expression, for precision gene delivery targeting.

University of Oxford Ludwig Institute for Cancer Research, Oxford, United Kingdom

Master's Student (Prof. Yang Shi), Sep 2021 – Jun 2022

- Executed genome-wide CRISPR-Cas9 knockout screening and computational hit prioritisation in cancer cell model, integrating NGS data analysis with experimental design.
- Identified and prioritized novel genetic vulnerabilities in pediatric glioma model, with top identified therapeutic target (ASCL1) advanced to *in vivo* validation via mouse models.

SKILLS

Statistical Genetics: GWAS (*REGENIE*, *PLINK*, *SNPTEST*), fine-mapping (*FINEMAP*), colocalisation (*coloc*), Mendelian randomisation (*TwoSampleMR*), PRS (*PRS-CS*, *SBayesRC*), LDSC regression, rare-variant association analysis (*REGENIE*, *SAIGE-GENE+*, *VEP*)

Multi-Omics: Proteomics (*Olink*), scRNA-seq/CITE-seq (*scanpy*, *scVI*, *muon*, *cellxgene*), bulk RNA-seq (*DESeq2*)

Programming & Infrastructure: Python (*pandas*, *NumPy*, *seaborn*, *scikit-learn*, *scikit-survival*), R (*tidyverse*, *ggplot2*), Bash, Git, HPC (*SLURM*), Cloud (GCP, *DNAexus*)

Pipelines & Reproducibility: Snakemake, WDL, Docker

Clinical Data Analysis: Survival/time-to-event analysis (Cox regression, Kaplan-Meier, RSF, GBST), systematic review and meta-analysis (*meta*)

Patient/Population Datasets: UK Biobank, All of Us, HCM Registry

PUBLICATIONS/PRESENTATIONS

Chan, J.H., Grace, C., Mahidi, M., Clarke, R., Ho, C., Neubauer, S., Kramer, C., Watkins, H., Goel, A.
Biobank-Scale Plasma Proteomics Identifies Novel Biomarkers in Hypertrophic Cardiomyopathy
Circulation: Precision & Genomic Medicine (*In Press*)

Chan, J.H., ..., Neubauer, S., Kramer, C., Watkins, H., Goel, A. on behalf of HCMR Investigators
Genotype-Phenotype Correlations Identify Clinically Significant Phenotypic Differences in Sarcomere Mutation-Positive Hypertrophic Cardiomyopathy in the NHLBI HCM Registry
Journal of the American College of Cardiology (*Under Review*)

Jiao, A.L., Maghrouni, A., Velazquez, A.G., Annett, A., Wen, T., Chan, J. ... Shi, Y.
H3K27M co-opts ASCL1 to maintain progenitor states and drive gliomagenesis
Nature Communications (*Under Review*)

Prioritising modifiable risk factors and biomarkers for hypertrophic cardiomyopathy: a Mendelian randomization approach
American Society of Human Genetics Annual Meeting (Boston, US), Oct 2025

Uncovering how Sarcomeric Variants affect Disease Heterogeneity in Hypertrophic Cardiomyopathy: A Cardiac Magnetic Resonance Imaging Study in the HCM Registry
International Society of Heart Research XXV World Congress (Nara, Japan), May 2025

Integrating Rare and Common Genetics for Risk Prediction in Hypertrophic Cardiomyopathy
Genomics England Research Summit (London, UK), Jul 2024

PROFESSIONAL SERVICE

Peer Review: Nature Cardiovascular Research, Circulation: Heart Failure, Nature Communications

President, Oxford University Biotech Society (Jun 2023 – Jun 2024): Managed team of 11 and spearheaded inaugural OxBioHack hackathon in collaboration with Oxford CompSoc and OPMS.